

Multi-Object Tracking for Apple Counting in Orchards Using Stereo Vision

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Abstract—Quantitative fruit counting at the individual tree level is a fundamental requirement in data-driven precision agriculture and a critical input for yield estimation models. However, achieving consistent tracking and accurate detection of fruits in unstructured outdoor environments remains a challenging task due to factors such as occlusion, varying illumination and sensor noise. Moreover, conventional 2D image-based methods are prone to scale ambiguity and duplicate detections from multiple viewpoints. To overcome these challenges, this work presents a 3D fruit counting framework based on stereo cameras in which each detection is localized within a georeferenced spatial coordinate system using GPS and visual odometry. The fruits are detected using a YOLOv8x model to identify object positions within individual images. To enable tracking across multiple consecutive frames, two state-of-the-art algorithms, OC-SORT and ByteTrack, are benchmarked against each other. OC-SORT, with incorporated optical flow, achieved the best performance on the validation dataset, with HOTA = 0.59, MOTA = 0.52, and IDF1 = 0.67, while the detection model reached a $mAP_{0.5:0.95}$ = 0.56. The proposed fruit counting framework demonstrated promising performance on Elstar trees, yielding an estimated total counting error of 20 apples, corresponding to a RMSE of 6.24 % and an R^2 score of 0.96. Given that fruit count is a key parameter to yield prediction, these results support accurate yield mapping and informed precision orchard management.

Index Terms—Fruit Counting, Multi-Object Tracking, Visual Odometry, Yield Estimation, 3D-Mapping, Precision Agriculture.

I. INTRODUCTION

Accurate fruit yield estimation is a critical aspect of agricultural production, as it directly influences harvest planning, storage logistics, market pricing, and supply chain management. Among various agronomic indicators such as blossom density and fruit size, fruit count remains a key parameter for reliably estimating yield [1]. Currently, farmers depend on pre-harvest yield estimations generated by third party software or agricultural consultants. These estimations are usually derived from manual counts performed on a small subsample of trees, which are extrapolated to the entire orchard using regression models. However, such limited sampling can introduce significant errors, as fruit-bearing patterns vary within different

years and are strongly influenced by the growing location [2]. This variability highlights the challenge and importance of developing robust methods for fruit counting.

The fundamental problem of fruit counting in orchards can be decomposed into a sequence of subtasks, including the detection and tracking of individual fruits, followed by the unique assignment to the corresponding trees.

More recent advancements in computational hardware have enabled deep learning methods to effectively handle dynamic environments, making them well-suited for object detection tasks in orchards. In particular, YOLO-based models combine high detection accuracy with real-time performance due to their efficient single-stage detection design [3], [4], [5]. A preliminary study presented in [6] evaluated the performance of several modern object detection algorithms, including Faster R-CNN, YOLOv5, YOLOv8 and RT-DETR, in the context of apple detection. While the results show that YOLOv8 and RT-DETR achieve comparable performance in $mAP_{50:95}$, YOLOv8 achieved lower computational overhead due to its one-stage architecture.

Counting fruits from a single image can be efficient in some cases but often leads to errors, particularly when fruits are partially occluded or not visible due to camera perspective [6]. Therefore, accurate fruit counting depends on the reliable tracking of individual fruits across multiple frames [3]. This can be achieved using multi-object tracking (MOT) algorithms, which assign consistent identifiers (IDs) to objects over a set of consecutive images. Moreover, MOT algorithms rely on well-established methods such as Kalman filter and Hungarian algorithm and offer a computationally lightweight solution, making them ideal for real-time applications [6].

Automatic fruit-to-tree assignment is still in its early stages, primarily due to overlapping tree canopies and the need to integrate observations from both the front and rear viewpoints of tree rows. To address these issues, this work introduces a 3D fruit-to-tree assignment framework that integrates object detection with fruit-level MOT. Unlike prior approaches [7], [8], which integrated a LiDAR for 3D localization and tree

association, the proposed method mainly depends on a stereo camera setup that is cost-effective and easy to deploy. Assigning fruits in 3D space provides a robust solution by addressing limitations of 2D image analysis, such as perspective-induced geometric distortion and the potential for duplicate counting of the same fruits from opposing sides. To enable consistent fruit tracking, OC-SORT [6], known for its improved robustness through recent observation history, alongside the widely used ByteTrack algorithm [9], which achieved the best performance for apple tracking in [3] are examined. To further mitigate the limitations of linear motion assumptions, optical flow is incorporated to provide more accurate velocity estimates and enhance the precision of state updates.

II. EXPERIMENTAL SETUP

The fruit-growing region along the southern banks of the Elbe River, spanning Hamburg and Lower Saxony, constitutes the largest contiguous area of fruit cultivation in Northern Europe. In this study, the dataset used for fruit counting was collected in the experimental test fields of the Esteburg, a regional research and advisory institution located in the Altes Land near Hamburg.

A. Hardware Setup

To support the development and validation of automated fruit counting, a multi-sensor carrier platform was developed in the context of a collaborative research initiative. As illustrated in Fig. 1a, the base configuration of the sensor setup is mounted on a robust steel frame, which can be attached to the standard three-point hitch of narrow-gauge tractors. Although a LiDAR sensor is part of the framework, it is currently only used for validation purposes, as a full integration into the processing pipeline remains impractical due to high associated costs. Image acquisition is carried out using two Basler ace 2 Pro industrial cameras (a2A5328-15ucPRO) in a custom stereo setup, each featuring a resolution of 24.4 MP and supporting frame rates of up to 15 fps. Given a maximum canopy height of 3 m and a fixed camera-to-tree distance of approximately 1.5 m, the cameras are equipped with Fujinon 1.2" lenses (CF8ZA-1S). Accurate georeferencing of the captured images is ensured by the simpleRTK3B Heading module with the mosaic-H chip from Septentrio, connected to two multiband GNSS antennas. The central computing unit is enclosed in a protective orange case and includes an NVIDIA Jetson AGX Orin Developer Kit, delivering the computational power required for real-time data processing and sensor integration. However, model training and evaluation were carried out on a high-performance workstation equipped with an NVIDIA RTX 6000 Ada GPU to accelerate computation and reduce processing time. A more detailed description of the sensor platform is provided in [10].

B. Data Collection

The image data were collected in different orchard sections of the experimental test fields at ESTEBURG. To facilitate the training of the detection algorithm, four apple cultivars were



Fig. 1: (a) Multi-sensor carrier platform. (b) Example image of Elstar tree. (c) Example image of Wellant tree.

selected: Elstar, Wellant, Rockit, and Pia 41. These cultivars exhibit inherent variations in fruit size and coloration, while image acquisition at different times of the day introduced a range of natural lighting conditions. Fig. 1b,c illustrates representative images of a Wellant and an Elstar tree used in this study. The image acquisition was carried out on 22 August for Elstar and on 5 September 2024 for the remaining cultivars, with all images captured shortly before harvest. During data acquisition, the tractor maintained an average speed of 1 m s^{-1} , while both cameras captured images with a resolution of 4608×5328 at a frame rate of 5 fps. To ensure accurate spatial alignment, the stereo camera setup was calibrated in the field prior to data acquisition using a PuzzleBoard design featuring positional encoding [11]. A total of 200 images were annotated with bounding boxes for object detection, while 131 consecutive frames of Wellant and Elstar were labeled with trajectories for the tracking task. For the final validation of fruit counts, data were collected from 30 Elstar apple trees within the experimental field. All annotations were performed using the CVAT (Computer Vision Annotation Tool) data annotation platform.

III. METHOD

A high-level schematic of the software framework is presented in Fig. 2. As previously mentioned, the apple counting can be inherently divided into the object detection and tracking task, which are handled by the computer vision module, and the subsequent matching of apples to trees. Since the assignment is performed in 3D space using only camera data,

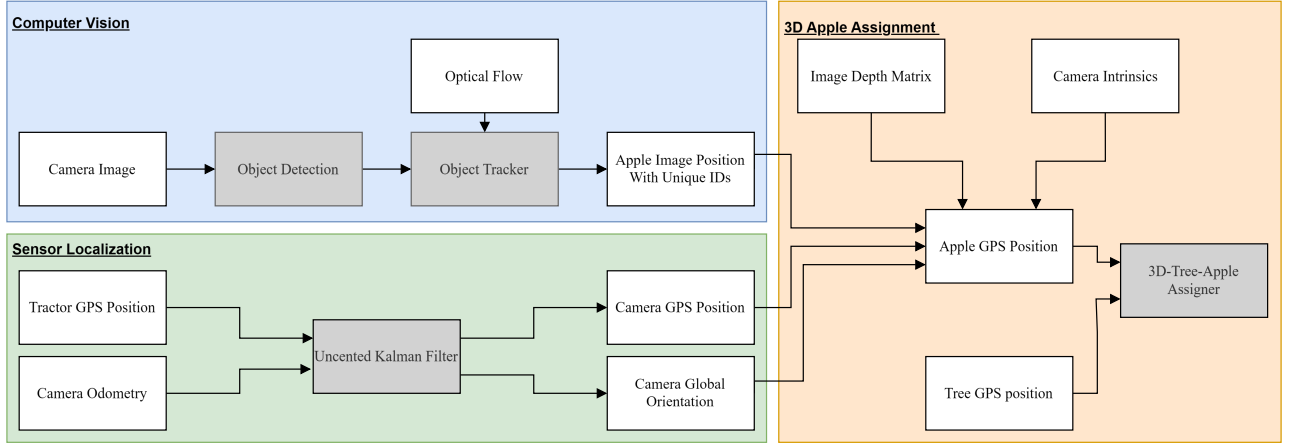


Fig. 2: Block diagram of the software framework, highlighting the interaction between functional components.

the detected apple positions have to be projected from the image plane into a globally referenced coordinate system. Therefore, the RTK-GNSS provides a suitable reference frame. In this context, an additional module is introduced to localize the sensor pose.

The following notation is adopted for clarity in the description of 3D projection: G denotes the global frame, C the camera frame and I the 2D image plane, such that $\mathbf{R}^{GC}$ represents a rotation from the global to the camera coordinate system.

A. Sensor Localization

In order to estimate the geometry of the scene and the motion of the camera in an orchard environment, a self-supervised deep visual stereo odometry (DVSO) algorithm [12] is employed to process synchronized stereo image pairs. The DVSO generates optical flow, dense depth maps and camera odometry. The optical flow is predicted using a deep learning model based on a customised PWC-Net [13], which analyses the displacement of image features between temporally adjacent images. For a well-rectified stereo camera setup, the relative transformation between the cameras can be considered constant due to the fixed sensor configuration. Under this assumption, the depth z of a point can be computed from the disparity d , the stereo baseline b , and the focal length f . The disparity is defined as the difference in horizontal image coordinates between the corresponding points in the left u_l and right u_r stereo image and can be inferred from the previously described flow estimation:

$$d = u_l - u_r = f \cdot \left(\frac{x - b}{z} - \frac{x}{z} \right) \Rightarrow z = -f \cdot \frac{b}{d}. \quad (1)$$

The camera ego-motion, including translation and rotation, is estimated from the optical flow between consecutive frames of a single camera and its corresponding depth map. While classical geometry-dependent methods such as the eight-point algorithm can be applied, they are sensitive to flow inaccuracies. Therefore, an additional neural network architecture is used to determine camera odometry. For the trainings

pipeline, the system leverages photometric consistency and 3D geometric constraints to learn without requiring annotated data. This enables rapid collection and training on vast data quantities, ensuring robustness under the challenging conditions of orchards.

Transforming the detected apples from the 2D image plane into the 3D reference frame requires knowledge of the position and orientation of the camera expressed in the global GNSS coordinate system. Consequently, the ego-motion estimated by the visual odometry algorithm must be aligned with a global coordinate system. Since this transformation involves rotations and translations between local and global frames, this process exhibits nonlinearity. To address this, an Unscented Kalman Filter (UKF) [14] is employed to fuse measurements from the GNSS sensor and the camera's visual odometry. The filter infers the global position and orientation of the camera as hidden states by combining local linear and angular velocity measurements from visual odometry and GNSS observations. Additionally, the GNSS observations correct accumulated drift and serve to anchor the estimates by providing georeferenced data. As a result, the global position vector $\mathbf{p}_{cam}^G$ and rotation matrix $\mathbf{R}^{GG'}$ of the camera can be determined, where GG' denotes the relative rotation between two consecutive camera poses in the georeferenced global frame.

B. Computer Vision

To detect apples in the image sequences, a YOLOv8x model [15] is employed. The model was initially pretrained on the COCO dataset and subsequently fine-tuned on the self-collected dataset described in Section II-B. Apple annotations were formatted as (x_c, y_c, w, h) to match YOLO's input requirements. In order to expand the training dataset and improve model robustness, image augmentations were applied using the Albumentations library, which enables detailed control over augmentation types and their respective probabilities. Additionally, hyperparameter tuning was performed using the Ray Tune library, applying random search to optimize the parameter configuration.

In the context of fruit tracking, the performance of two

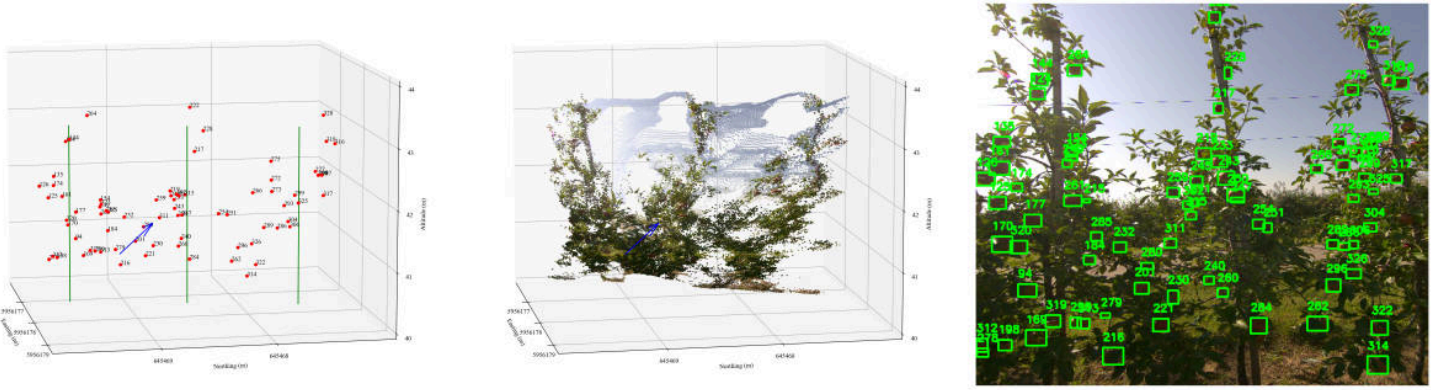


Fig. 3: Projection of a 2D image segment containing Wellant trees into 3D space using stereo vision.

tracking algorithms, ByteTrack and OC-SORT, is evaluated based on their capability to consistently preserve unique object IDs across consecutive image frames. Both tracking algorithms incorporate the Kalman Filter, which predicts object locations with motion estimates from previous frames. ByteTrack implements a two-stage association process using the Hungarian algorithm. In the first stage, high-confidence detections are matched with predicted positions using IoU. In the second stage, unmatched tracks are further associated with low-confidence detections to compensate for missed associations. While ByteTrack focuses on tracking accuracy, OC-SORT improves robustness under occlusion by incorporating a motion consistency term into a cost matrix of the Hungarian algorithm. The object's velocity is estimated from recent positional changes and used to evaluate how closely new detections in the current frame align with its expected motion, forming the motion similarity cost. Given that optical flow is estimated by DSVO, the impact of incorporating velocity as an additional measurement and modeling its corresponding measurement noise based on certainty values is integrated into the Kalman filter.

C. 3D Apple Assignment

The total number of fruits per tree is estimated by measuring the proximity of apples using the euclidean distance. Due to the dense canopies of some apple varieties, object detection struggles to reliably identify trees, even when trunks are partially visible. To address this, the positions $\mathbf{p}_{t_j}^G$ of each tree j are recorded separately within a row, using a GNSS-mounted pole. Since tree positions remain constant over the years, they only need to be manually registered once. Alternatively, trees could be localized with object detection during winter, when the absence of foliage reduces visual obstruction. Associating apples with individual trees requires reprojecting image-plane detections into the globally referenced 3D space. As a first step, the center coordinates of each detected apple i denoted as $\mathbf{p}_{a_i}^I = (x_c, y_c)$ are mapped from 2D image space into 3D camera space using the camera's intrinsic matrix $\mathbf{K}$

and the average depth $\bar{z}$, computed over the number of pixels N within a bounding box:

$$\mathbf{p}_{a_i}^C = \bar{z} \cdot \mathbf{K}^{-1} \begin{bmatrix} x_c \\ y_c \\ 1 \end{bmatrix}, \quad \text{where } \bar{z} = \frac{1}{N} \sum_{i=1}^N z_{\text{bbox}_i}. \quad (2)$$

Subsequently, the objects represented in the 3D camera coordinate system are transformed into the global coordinate system by using the rotation matrix $\mathbf{R}^{GG'}$ and translation vector $\mathbf{p}_{\text{cam}}^G$, derived in Section III-A. Additionally, the extrinsic rotation between the camera and the reference GNSS coordinate system is accounted with $\mathbf{R}^{GC}$:

$$\mathbf{p}_{a_i}^G = \mathbf{R}^{GG'} \cdot \mathbf{R}^{GC} \cdot \mathbf{p}_{a_i}^C + \mathbf{p}_{\text{cam}}^G. \quad (3)$$

With both apple positions $\mathbf{p}_{a_i}^G$ and tree positions $\mathbf{p}_{t_j}^G$ expressed in the GNSS-referenced coordinate system, apples can be assigned to the trees based on Euclidean distance:

$$j_{\min} = \arg \min_j \left\| \mathbf{p}_{a_i}^G - \mathbf{p}_{t_j}^G \right\|_2. \quad (4)$$

This procedure is repeated for each detected apple within a frame to identify the tree index with the smallest distance $j_{\min}$, allowing for the assignment of the total fruit counts. Potential duplicate counts are mitigated using unique IDs generated by the object tracker. Fig. 3 illustrates an example of projecting apples into 3D space using the proposed reconstruction approach.

IV. RESULTS

The detection performance is measured using $\text{mAP}_{0.5:0.95}$, which captures both localization and classification accuracy across multiple IoU thresholds. The tracking performance is evaluated using HOTA, MOTA, and IDF1, where HOTA reflects the joint accuracy of detection and association, MOTA emphasizes overall object coverage and penalizes identity switches, and IDF1 measures the consistency of identity preservation. To ensure an unbiased evaluation, the training dataset for the object detection algorithm was selected to

have no overlap with the testing dataset used for the results presented in this chapter. Following this setup, the YOLOv8x architecture achieved a performance of $mAP_{0.5:0.95} = 0.56$ using k-fold cross-validation, with the dataset split into 80% for training and 20% for validation. All visible apples, including those with more than 90% occlusion, were annotated. This resulted in a challenging detection scenario due to limited visual cues. However, the obtained F1 score of 0.78 reflects the model's ability to maintain a balanced precision-recall trade-off. Overall, the performance is consistent with results reported in [4], [16], [17].

The evaluation results of the object tracking algorithms are shown in Table I. The Elstar and the Wellant datasets were chosen specifically for their contrasting visual and structural characteristics, allowing for a systematic comparison of the trackers performance across distinct configurations. The Wellant dataset features a spindle-trained tree structure with high fruit density and relatively low levels of occlusion, whereas the Elstar dataset exhibits a dense canopy with fewer apples and severe occlusions. Among the evaluated methods that integrate optical flow, OC-SORT achieved the highest HOTA score of 0.64 across all datasets.

TABLE I: Tracking metrics for ByteTrack and OC-SORT with and without optical flow on Wellant and Elstar

Tracker	Opt. Flow	Dataset	HOTA	MOTA	IDF1	DeTA
ByteTrack	No	Wellant	0.59	0.62	0.66	0.63
	No	Elstar	0.58	0.51	0.67	0.50
	Yes	Wellant	0.63	0.63	0.70	0.63
	Yes	Elstar	0.58	0.52	0.67	0.50
OC-SORT	No	Wellant	0.61	0.68	0.74	0.64
	No	Elstar	0.59	0.52	0.67	0.52
	Yes	Wellant	0.64	0.64	0.70	0.65
	Yes	Elstar	0.59	0.53	0.69	0.51

In general, a Kalman filter tracker instantiate a separate Kalman filter instance for every individual object. Each filter remains active throughout the lifespan of the tracked object and is terminated once the object can no longer be reliably associated. During this period, the filter incrementally refines its state estimates through recursive updates based on incoming measurements. However, when the image acquisition frequency is low, the filter receives fewer updates, weakening its ability to accurately learn motion dynamics. Consequently, at an image acquisition rate of 5 fps, the effect of random initialization becomes more significant, resulting in degraded tracking performance. In this experiment, the tracker without optical flow still achieved similar results, primarily because the initial velocity estimates were pre-set through hyperparameter optimization using Ray Tune. While effective, this approach requires manual reconfiguration for each dataset or an additional mechanism to infer initial velocities from external sensors. Failing to do so leads to a substantial decline in robust tracks, as evidenced in [18]. The performance difference across datasets suggest that Elstar poses a substantially more challenging detection scenario than Wellant, as indicated by the lower DeTA scores. Since OC-SORT with optical flow

integration achieved the best tracking performance, it was adopted for the following fruit count validation.

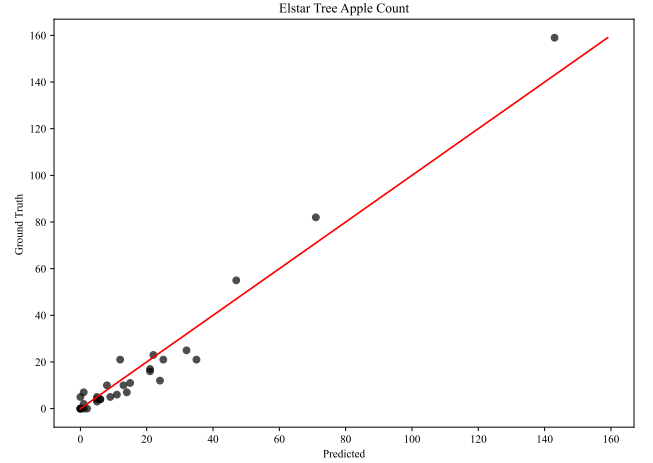


Fig. 4: Scatter plot comparing ground truth against estimated apple counts of 30 Elstar trees.

For the validation experiment, the Elstar cultivar was exclusively selected, as it was the only variety harvested using the Pluk-O-Trak platform (Munckhof Fruit Tech Innovators) in 2024, enabling the acquisition of machine-counted fruits. This choice was made to minimize the potential human bias introduced by hand counting fruits. Based on the collected data, fruit counts were obtained from both sides of each tree and subsequently matched using their global GPS coordinates. A distance threshold was defined based on pre-harvest diameter of Elstar apples, assumed to range from 70 to 85 mm. Apples positioned closer to each other than the defined threshold were considered duplicate detections and therefore one instance for each duplicate was removed during the matching process. A scatter plot of the counting result is presented in Fig. 4, where the ground truth is plotted against the estimated counts per tree. The red line represents the line of perfect agreement between the estimates and the actual counts. Using the Pluk-O-Trak, a total of 535 fruits were machine-counted across the 30 sampled Elstar trees. The counting method proposed in this study initially estimated 619 apples, which was reduced to 555 after applying the threshold. This corresponds to a total error of 20 apples, with a root mean square error (RMSE) of 6.24 % per tree, indicating low average deviation, and a coefficient of determination (R^2) of 0.96, reflecting overall strong agreement with ground truth counts. Due to the dense canopy structure, many apples were only visible from one side of the tree row, resulting in a smaller impact of duplicate detections. One of the main contributing factors to the overestimation in total apple count is assumed to be the occurrence of identity switches, arising from limitations in the tracker's ability to consistently associate detections over time as indicated by the IDF1 score of 0.69 in Table I. Additionally, the inclusion of green PIA41 apples in the training dataset increased the likelihood of false positive detections, as their visual similarity to small leaves can lead to misclassifications.

Overall, the proposed framework demonstrated reasonably ac-

curate fruit counts, highlighting the effectiveness of associating individual fruits to trees in 3D space.

V. CONCLUSION AND OUTLOOK

In this work, a 3D vision-based apple counting framework was developed to enable the association of fruits with individual trees. A custom dataset was collected using a multi-sensor carrier platform within an experimental orchard environment to train and evaluate the object detection and tracking components.

Specifically, a YOLOv8x model was used and trained on images of multiple different apple cultivars to ensure robust detection performance across varying fruit characteristics. The resulting detections served as input to the tracking pipeline, where two state-of-the-art algorithms, ByteTrack and OC-SORT, were evaluated and compared. Given that the data were collected at a high resolution and a low frame rate of 5 fps, optical flow was incorporated to support motion prediction in the Kalman filter-based tracker. As a result, OC-SORT with optical flow outperformed the other methods across all benchmarking metrics. Moreover, reliable performance without optical flow is highly dependent on correctly initialized states, particularly the object's velocity.

To mitigate 2D image artifacts such as perspective distortion and to address the issue of double counting fruits observed from both sides of a tree row, each tracked fruit was projected into 3D space. Global fruit positions were estimated using an UKF, incorporating GPS data and camera-based odometry. The proposed counting framework yielded a RMSE of 6.24% per tree and a $R^2 = 0.96$, indicating a strong statistical fit. For future work, additional data must be collected and annotated to thoroughly evaluate the generalizability of the framework. In particular, access to 3D-labeled fruit data from LiDAR would enable the development and evaluation of tracking methods operating entirely in 3D space, potentially reducing or eliminating reliance on 2D trackers.

Ultimately, robust and precise fruit counting serves as a foundational step for downstream applications in orchard management. Accurate fruit counts are critical for size estimation, growth monitoring and yield prediction, which informs logistics and labor planning. Furthermore, per-tree count data can support precision interventions such as targeted canopy management and nutrient optimization, while also providing insights into bearing cycles. As orchard systems move increasingly toward automation and data-driven decision-making, scalable and accurate fruit counting solutions will become an essential component of modern horticultural practice.

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